

HIGH INTENSITY TRAINING (HIT) INFOGRAPHICS COLLECTION

 knowledgeiswatt.substack.com/p/high-intensity-training-hit-infographics

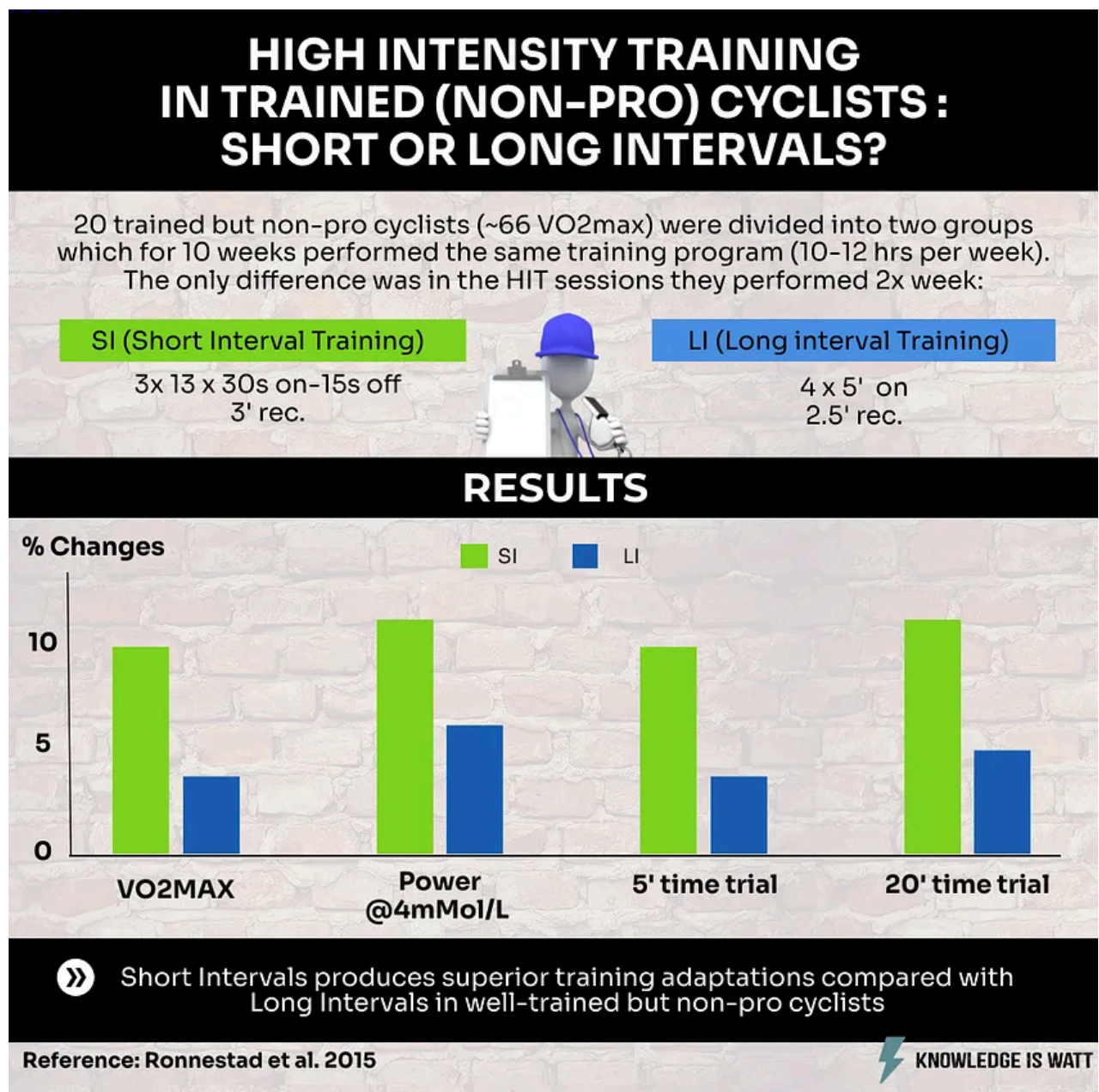
Knowledgeiswatt

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Here is the growing infographics library about all the relevant researches on **high intensity training and cycling performance** with links to each study's summary.

1. SHORT VS LONG HIT INTERVALS IN WELL-TRAINED (NON-PRO) CYCLISTS:

[Research Summary and Comment](#)



2. SHORT VS LONG HIT INTERVALS IN PRO CYCLISTS

[Research Summary and Comment](#)

SHORT VS LONG HIGH INTENSITY INTERVALS IN PRO CYCLISTS

18 elite cyclists were divided into two groups and performed 3 times per week for 3 weeks two different interval training sessions performed at the maximum sustainable intensity (max effort).

SI (Short Interval Training)

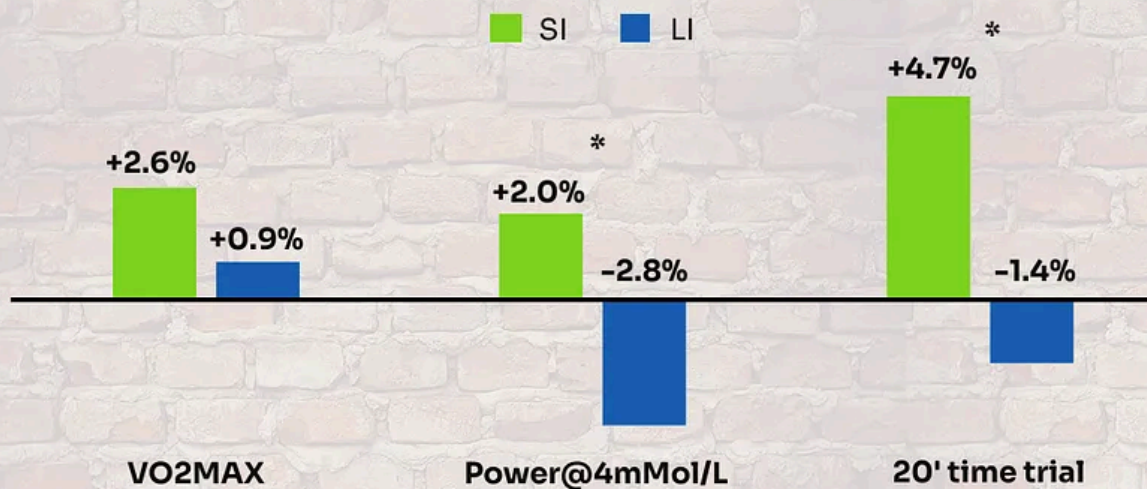
3x 13 x 30s on-15s off
3' rec.



LI (Long interval Training)

4 x 5' on
2.5' rec.

RESULTS



Short high intensity intervals produce superior training adaptations compared to Long Intervals in professional cyclists

Reference: Ronnestad et al. 2020

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3. LONG HIT INTERVALS: FAST START VS STEADY PACE

[Research Summary and Comment](#)

FAST-START LONG HIT INTERVALS

11 well-trained cross-country skiers ($\text{VO}_{2\text{max}}$ 70) completed 2 different HIT (High Intensity Training) sessions. MAP is Maximal Aerobic Power.

FAST START LONG INTERVALS

- 5x5min as follows:
- The first 1min30s at 100% MAP
 - Then, 3min30s at 85% MAP
3min rec.



STEADY PACE LONG INTERVALS

5x5min at 90% of MAP with
3min recovery.



RESULTS

Time
> 90%
 $\text{VO}_{2\text{MAX}}$



RPE



» In fast start long intervals, a higher fraction of $\text{VO}_{2\text{max}}$ is sustained, despite a lower RPE (perception of effort).
This could help to maximize performance adaptations

Rønnestad et al. 2020



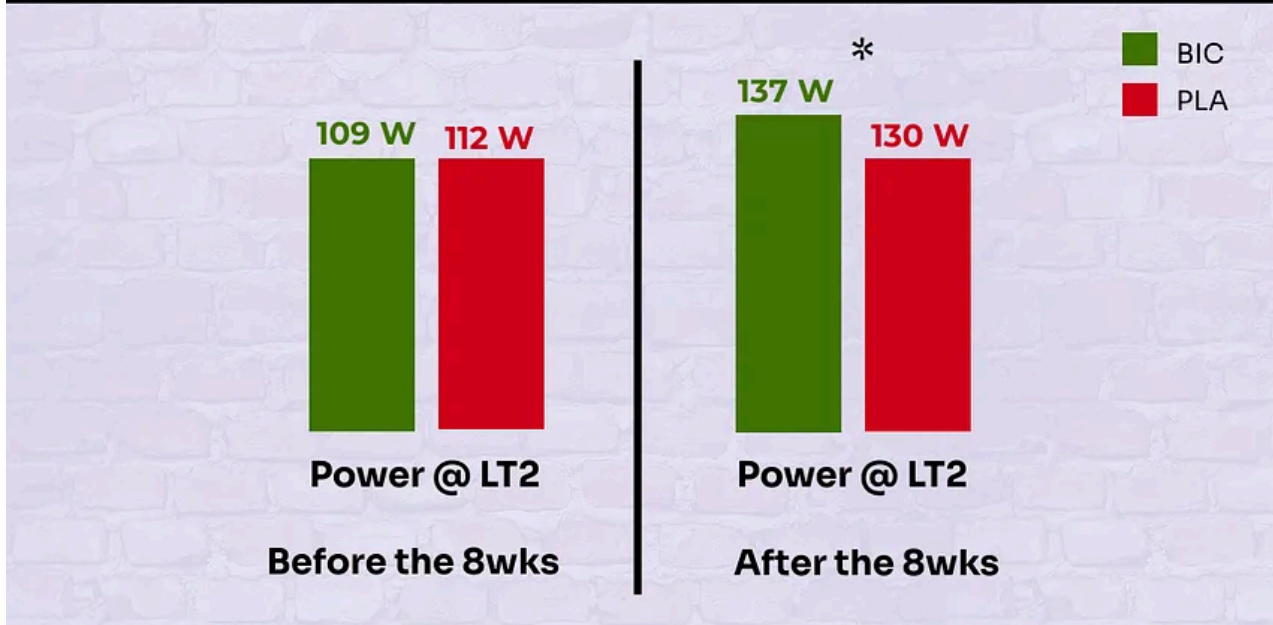
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4. BICARBONATE BEFORE HIT SESSIONS

[Research Summary and Comment](#)

BICARBONATE BEFORE HIT TRAINING BOOSTS PHYSIO ADAPTATIONS

16 recreationally active women were placed into the BIC (n = 8) or PLA (n = 8) groups, and performed 8 wk (3 days/wk) of 6-12 X 2min cycling intervals at 140-170% of their second lactate threshold (LT2) (HIT, High Intensity Training). They ingested Bicarbonate (0.4 g/kg) or a placebo before each training session



Bicarbonate supplementation before High Intensity Training can enhance aerobic adaptations



This because reducing metabolic acidosis (H⁺) during HIT could facilitate biological signalling to improve muscle oxidative capacity.

Reference: Edge et al 2006



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LITERATURE BRIEFING:

- Both in well-trained amateur and professional cyclists, short HIT intervals (3x13min alternating 30s on and 15s off) seems to provide superior physiological and performance adaptations compared to Long HIT intervals (4x5min with 2.5min rec.) when both formats are performed at the maximum sustainable intensity. (1, 2)
- When considering long HIT intervals, a fast start for each interval (1m30s at 100% VO₂max power followed by 3m30s at 85% of VO₂max power) permits to spend more time at a high percentage of VO₂max (higher aerobic load) with a lower perceived effort compared to maintain a steady pace (all 5min at 90% of VO₂max power). (3)
- Ingesting sodium bicarbonate (0.4 g/kg) before HIT sessions for some weeks can boost training adaptations and performance. (4)

REFERENCES:

1. Rønnestad BR, Hansen J, Vegge G, Tønnessen E, Slettaløkken G. Short intervals induce superior training adaptations compared with long intervals in cyclists - an effort-matched approach. *Scand J Med Sci Sports*. 2015 Apr;25(2):143-51.
2. Rønnestad BR, Hansen J, Nygaard H, Lundby C. Superior performance improvements in elite cyclists following short-interval vs effort-matched long-interval training. *Scand J Med Sci Sports*. 2020 May;30(5):849-857.
3. Rønnestad BR, Rømer T, Hansen J. Increasing Oxygen Uptake in Well-Trained Cross-Country Skiers During Work Intervals With a Fast Start. *Int J Sports Physiol Perform*. 2020 Mar 1;15(3):383-389.
4. Edge J, Bishop DJ, Goodman C. Effects of chronic NaHCO₃ ingestion during interval training on changes to muscle buffer capacity, metabolism, and short-term endurance performance. *J Appl Physiol* 101: 918–925, 2006.